Exercise 2: Uncertainty Prediction

This document shows the implementation of error estimation for a CNN model trained on the GALAH star spectra dataset. The model was used to predict the three labels t_eff, log_g, and fe_h.

Result Plots

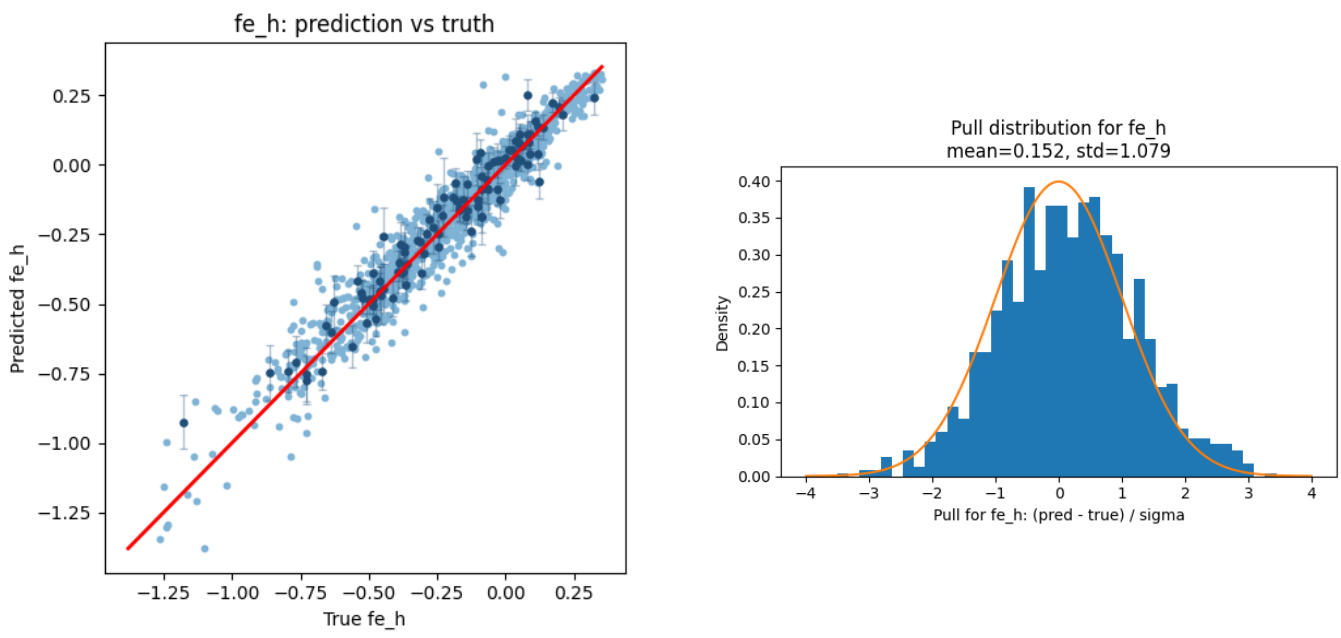

Figure 1: Results for metallicity, [Fe/H]. Left: predicted versus true values with a subset of values with representative uncertainty bars and the red diagonal indicating perfect agreement. Right: pull distribution for [Fe/H], compared to the standard normal distribution to assess uncertainty calibration.

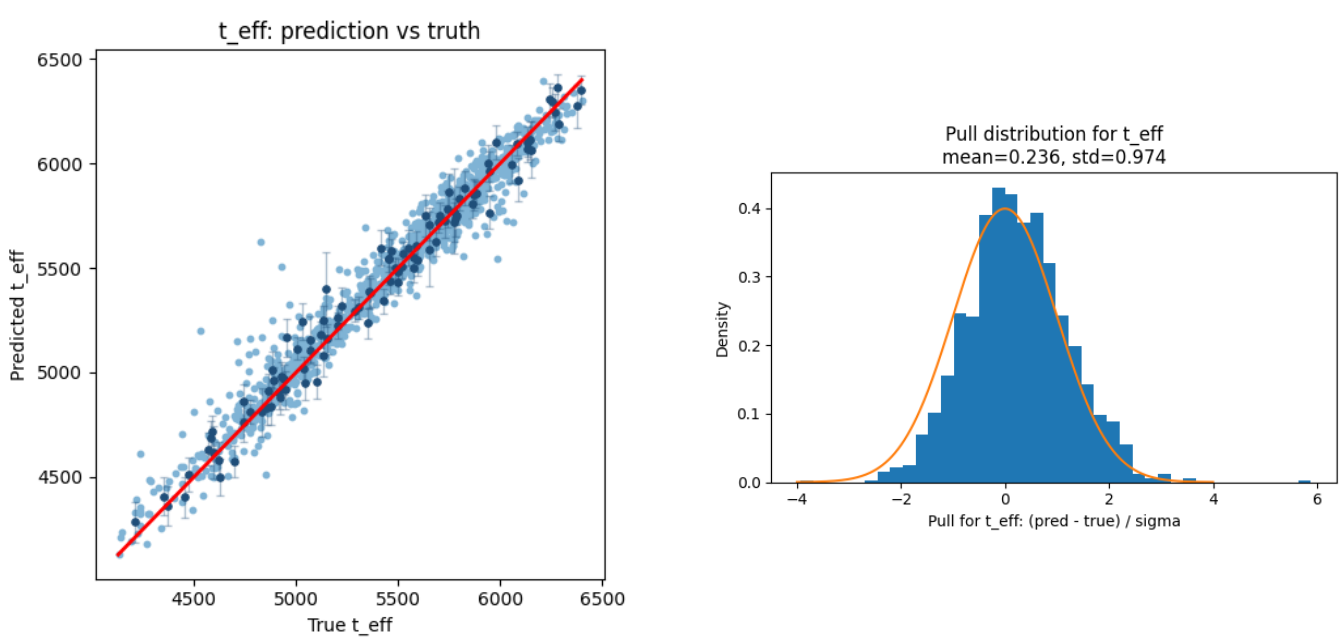

Figure 2: Results for effective temperature, T_{eff}. Left: predicted versus true values with a subset of values with representative uncertainty bars and the red diagonal indicating perfect agreement. Right: pull distribution for T_{eff}, compared to the standard normal distribution to assess uncertainty calibration.

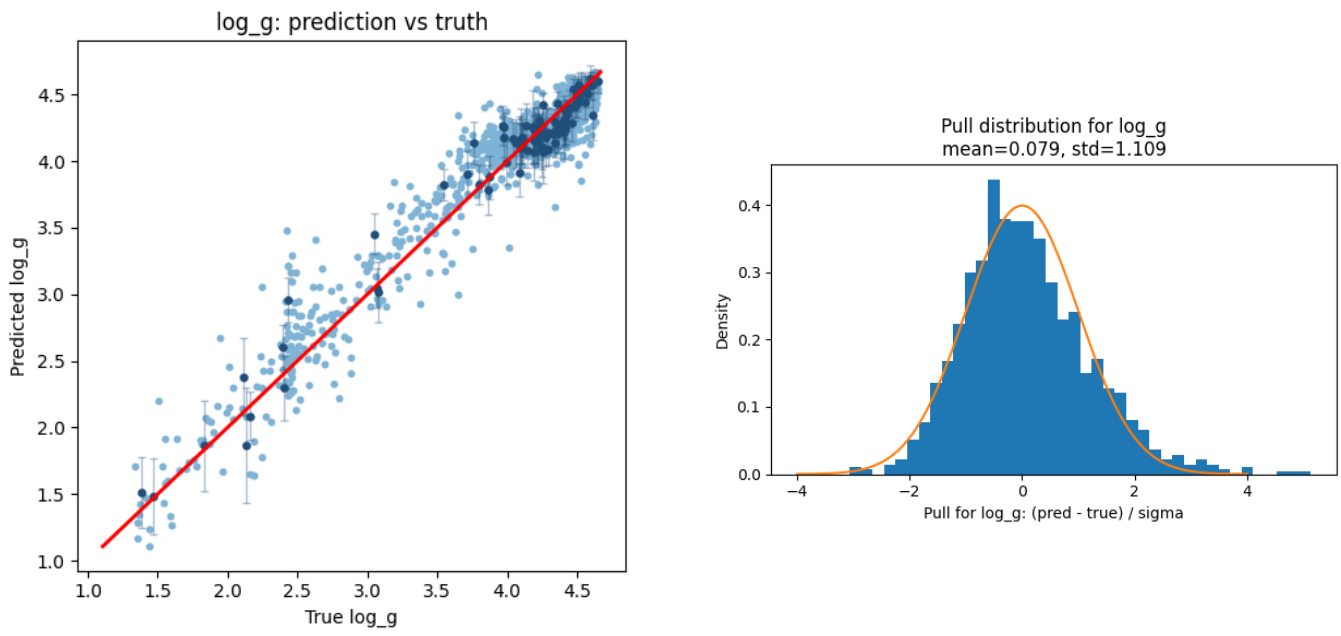

Figure 3: Results for surface gravity, $\log g$. Left: predicted versus true values with a subset of values with representative uncertainty bars and the red diagonal indicating perfect agreement. Right: pull distribution for $\log g$, compared to the standard normal distribution to assess uncertainty calibration.

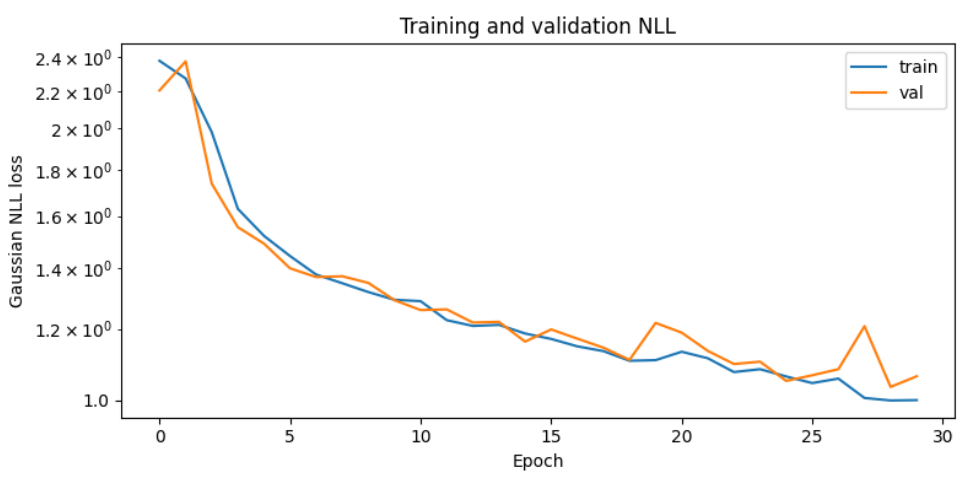

Figure 4: Training and validation NLL (negative log-likelihood) during training. Both curves decrease in a similar way, which suggests stable learning without strong overfitting.